Teacher Answer Key

Mechanical Oscillations - Exercise 2: Potential Energy

1. Coefficient 'a' and Mechanical Energy

a) Calculate constant a:

The potential energy is given by the parabolic function $EPE = ax^2$.

From the graph, we identify a clear reading point at the maximum elongation (amplitude):

For $x = 3 \text{ cm} = 0.03 \text{ m}$, let's read the corresponding energy $EPE = 9 \times 10^{-2} \text{ J}$.

$$a = \frac{EPE}{x^2} = \frac{9 \times 10^{-2}}{(3 \times 10^{-2})^2}$$

$$a = \frac{9 \times 10^{-2}}{9 \times 10^{-4}} = \frac{10^{-2}}{10^{-4}} = 100$$

Unit of a: Since EPE is in Joules and x in meters, a is in J/m^2 (or N/m).

Result: $\boxed{a = 100 \text{ J/m}^2}$

b) Expression of Mechanical Energy (ME):

The system comprises Kinetic Energy (KE) and Elastic Potential Energy (EPE). $GPE = 0$.

$$ME = KE + EPE$$

$$KE = \frac{1}{2}Mv^2 \quad \text{and} \quad EPE = ax^2$$

Substituting $M = 400 \text{ g} = 0.4 \text{ kg}$ and $a = 100$:

$$ME = \frac{1}{2}(0.4)v^2 + 100x^2$$

Expression: $\boxed{ME = 0.2v^2 + 100x^2}$ (in SI units)

2. Differential Equation & Period

Since the oscillator is free and un-damped, the Mechanical Energy is conserved.

Law of Conservation of Mechanical Energy:

$$\frac{dME}{dt} = 0$$

$$\frac{d}{dt}(0.2v^2 + 100x^2) = 0$$

$$0.2(2v \cdot v') + 100(2x \cdot x') = 0$$

Knowing that $x' = v$ (since RM) and $v' = x''$:

$$0.4vx'' + 200xv = 0$$

Dividing by v (where $v \neq 0$):

$$0.4x'' + 200x = 0 \implies x'' + \frac{200}{0.4}x = 0$$

Differential Equation: $\boxed{x'' + 500x = 0}$

Deduce the Period (T_0):

The equation is of the form $x'' + \omega_0^2 x = 0$, with $\omega_0^2 = 500$.

$$T_0 = \frac{2\pi}{\omega_0} = \frac{2\pi}{\sqrt{500}} = \frac{2\pi}{10\sqrt{5}}$$

$$T_0 = \frac{\pi}{5\sqrt{5}} \approx \frac{3.14}{11.18}$$

Period: $\boxed{T_0 \approx 0.28 \text{ s}}$

3. Time Equation & Graph ($t = 0, x_0 = -1\text{cm}, v_0 < 0$)

a) Solution of Differential Equation:

General form: $x = X_m \sin(\omega_0 t + \varphi)$.

1. Find X_m: The motion is bounded by the graph limits $[-3\text{cm}, +3\text{cm}]$.

Thus, $\boxed{X_m = 3 \text{ cm} = 0.03 \text{ m}}$

2. Find ω_0: From part 2, $\boxed{\omega_0 = \sqrt{500} \approx 22.36 \text{ rad/s}}$

3. Find Phase Angle φ: At $t = 0$, $x = -1$ cm.

$$x(0) = X_m \sin(\varphi) \implies -1 = 3\sin(\varphi)$$

$$\sin(\varphi) = -\frac{1}{3}$$

Two possible angles in $[0, 2\pi]$:

$\varphi_1 \approx 3.48$ rad (3rd quadrant) or $\varphi_2 \approx 5.94$ rad (4th quadrant).

Check Velocity sign: $v(0) < 0$.

$$v = x' = X_m \omega_0 \cos(\omega_0 t + \varphi)$$

At $t = 0$: $v_0 = X_m \omega_0 \cos(\varphi)$. Since $X_m, \omega_0 > 0$, we need $\cos(\varphi) < 0$.

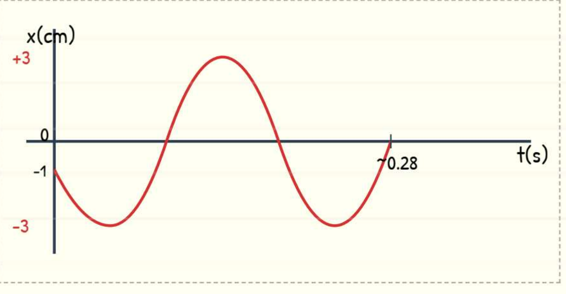
- In 3rd quadrant, $\cos(\varphi) < 0$ (Accepted).
- In 4th quadrant, $\cos(\varphi) > 0$ (Rejected).

Thus, $\varphi \approx 3.48$ rad (or $\pi + 0.34$ rad).

Final Equation: $\boxed{x = 0.03 \sin(22.36t + 3.48)}$ (SI)

b) Graph of $x = f(t)$:

The oscillator starts at $x = -1$ cm and moves in the negative direction (downwards/left) towards $-X_m = -3$ cm.



(Sketch: Starts at -1cm, reaches min -3cm quickly, then oscillates)